

Ready-Mix Concrete Technical Data & QC Logs

Mix Designs (M10–M40), Production Standards (IS 4926 / IS 10262), Quality Control Logs & API Import Schemas

1. RMC Mix Design Matrix (M10, M20, M25, M30, M40) - Per 1 m³

Standardized proportioning reports showing variations in cementitious content, aggregates, water demand, and superplasticizer dosages.

Grade / Parameter	M10 (PCC)	M20 (RCC Standard)	M25 (RCC Medium)	M30 (RCC High-Durability)	M40 (High Strength)
Primary Application	Levelling / PCC	Footings / Slabs	Beams & Columns	Heavy Structural / Pavements	High-Rise Columns / Pre-stressed
Characteristic Strength (f_{ck})	10 N/mm²	20 N/mm²	25 N/mm²	30 N/mm²	40 N/mm²
Target Mean Strength (f'_{ck})	13.5 N/mm²	26.5 N/mm²	31.6 N/mm²	38.2 N/mm²	48.2 N/mm²
Cement Content (OPC 53 Grade)	220 kg/m³	320 kg/m³	350 kg/m³	380 kg/m³	430 kg/m³
Fly Ash / Mineral Admixture	—	—	30 kg/m³ (Optional)	50 kg/m³ (PPC/Flyash)	70 kg/m³ (Micro Silica/FA)
Fine Aggregate (M-Sand / Zone II)	750 kg/m³	680 kg/m³	660 kg/m³	640 kg/m³	610 kg/m³
Coarse Aggregate (20 mm)	700 kg/m³	660 kg/m³	670 kg/m³	660 kg/m³	650 kg/m³
Coarse Aggregate (10 mm)	480 kg/m³	440 kg/m³	440 kg/m³	430 kg/m³	430 kg/m³
Free Water Content	130 L/m³	160 L/m³	155 L/m³	150 L/m³	145 L/m³
Water-Cement Ratio (w/c)	0.60	0.50	0.44	0.39	0.34
Superplasticizer Dosage	0.0%	0.6% (1.92 kg)	0.8% (2.80 kg)	1.0% (3.80 kg)	1.2% (5.16 kg)
Target Slump at Plant	100 ± 25 mm	120 ± 25 mm	140 ± 25 mm	150 ± 25 mm	160 ± 25 mm

2. Production Standards & Batching Plant Operating Guidelines (IS 4926)

A. Batching Tolerances & Mixing Standards

- **Cement & Cementitious Materials:** ±1.0% by weight
- **Water & Chemical Admixtures:** ±1.0% by weight/volume
- **Aggregates (Coarse & Fine):** ±2.0% by weight
- **Mixing Time:** Min 30 seconds for twin-shaft pan mixers; min 60 seconds for drum mixers.
- **Moisture Compensation:** Automatic real-time correction for aggregate surface moisture.

B. Transit & Discharge Parameters

- **Transit Drum Agitating Speed:** 2 to 6 RPM during transport.
- **Site Re-Mixing Speed:** 12 to 18 RPM for 3–5 minutes prior to discharge.
- **Max Time Limit to Discharge:** 2.0 hours (or 1.5 hours if ambient temp > 30°C).
- **Addition of Water on Site:** Strictly prohibited unless specified by the Quality Control Lead.

3. Sample Quality Control Logs & Test Records

Representative batch inspection and compressive cube testing log for software audit trail verification.

Batch ID	Date / Time	Grade	Truck ID	Slump (mm)	Temp (°C)	7-Day Strength	28-Day Strength	QC Status
#RMC-2026-0891	2026-08-15 08:30	M20	TM-04	115 mm	28.5 °C	14.8 N/mm² (74%)	22.1 N/mm² (110%)	PASS

Batch ID	Date / Time	Grade	Truck ID	Slump (mm)	Temp (°C)	7-Day Strength	28-Day Strength	QC Status
#RMC-2026-0892	2026-08-15 09:15	M25	TM-12	135 mm	29.1 °C	18.2 N/mm² (73%)	27.8 N/mm² (111%)	PASS
#RMC-2026-0893	2026-08-15 10:00	M30	TM-08	145 mm	30.4 °C	21.5 N/mm² (72%)	33.4 N/mm² (111%)	PASS
#RMC-2026-0894	2026-08-15 11:20	M40	TM-02	160 mm	31.0 °C	29.4 N/mm² (73.5%)	43.8 N/mm² (109.5%)	PASS
#RMC-2026-0895	2026-08-15 12:05	M10	TM-07	95 mm	31.5 °C	7.8 N/mm² (78%)	11.4 N/mm² (114%)	PASS

4. Master JSON Schema for Software Data Import

Use the following complete JSON schema to parse and load mix designs, standards, and QC logs into software database tables:

```
{
  "rmc_system_data": {
    "mix_designs": [
      { "grade": "M10", "fck": 10.0, "cement_kg": 220, "flyash_kg": 0, "fine_agg_kg": 750, "coarse_20mm_kg": 700, "coarse_10mm_kg": 480, "water_l": 130, "wc_ratio": 0.60, "admixture_kg": 0.0 },
      { "grade": "M20", "fck": 20.0, "cement_kg": 320, "flyash_kg": 0, "fine_agg_kg": 680, "coarse_20mm_kg": 660, "coarse_10mm_kg": 440, "water_l": 160, "wc_ratio": 0.50, "admixture_kg": 1.92 },
      { "grade": "M25", "fck": 25.0, "cement_kg": 350, "flyash_kg": 30, "fine_agg_kg": 660, "coarse_20mm_kg": 670, "coarse_10mm_kg": 440, "water_l": 155, "wc_ratio": 0.44, "admixture_kg": 2.80 },
      { "grade": "M30", "fck": 30.0, "cement_kg": 380, "flyash_kg": 50, "fine_agg_kg": 640, "coarse_20mm_kg": 660, "coarse_10mm_kg": 430, "water_l": 150, "wc_ratio": 0.39, "admixture_kg": 3.80 },
      { "grade": "M40", "fck": 40.0, "cement_kg": 430, "flyash_kg": 70, "fine_agg_kg": 610, "coarse_20mm_kg": 650, "coarse_10mm_kg": 430, "water_l": 145, "wc_ratio": 0.34, "admixture_kg": 5.16 }
    ],
    "production_standards": {
      "batching_tolerances_percent": { "cementitious": 1.0, "water_and_admixtures": 1.0, "aggregates": 2.0 },
      "transit_mixer_rpm": { "agitating": "2-6 RPM", "mixing": "12-18 RPM" },
      "max_discharge_time_hours": 2.0
    },
    "qc_logs": [
      { "batch_id": "RMC-2026-0891", "date": "2026-08-15T08:30:00", "grade": "M20", "truck_id": "TM-04", "slump_mm": 115, "temp_c": 28.5, "strength_7d_N_mm2": 14.8, "strength_28d_N_mm2": 22.1, "status": "PASS" }
    ]
  }
}
```